SISO Design Limitations – Structural Limitations

(SISO系统设计的局限性 – 结构限制)

Structural Limitations via Time Delays

- **Best Attainable Ideal Sensitivity Functions for a Delay of τ**

  (对时延τ 来说最佳可达到的理想灵敏度函数)

$$T_n(s) = e^{-s\tau} \qquad S_n^*(s) = 1 - e^{-s\tau}$$

To achieve the above ideal result requires the use of Smith Predictor plus an ideal controller！**(为达到上述理想的结果，需要使用Smith预测器和一个理想的控制器)**

- **Practically Achievable Sensitivity Functions(实际可达灵敏度函数)**

For ητ error of time delay, magnitude of MME $P_\Delta(s)$ approaches to 1 at approximately an order of 1/ητ, and thus limits the achievable CL bandwidth to this order for robust stability.**(对于时延为ητ的误差，$P_\Delta(s)$的大小以1 /ητ的数量级接近1，因此会将可获得的闭环带宽限制在此数量级，以实现鲁棒稳定性)**

(i)Delays limits disturbance rejection by requiring that a delay occurs before the disturbance can be cancelled. **(时延限制了干扰的抑制，要求时延在干扰能被抵消之前发生)**

(ii)Delays further limit the achievable CL bandwidth through the impact of modeling errors **(时延通过建模误差的影响进一步限制了闭环带宽的可达性)**

Effect of Plant Poles and Zeros

- **Zero Complementary Sensitivity Value at all Uncancelled Plant Zeros (所有未消去的系统零点处的零互补灵敏度值)**
  - The zeros of the complementary sensitivity function include all uncancelled plant and controller zeros (互补灵敏度函数的零点包括所有未消去的系统零点和控制器零点)

$$T(s)=\frac{L(s)}{1+L(s)}=\frac{P(s)C(s)}{1+P(s)C(s)}$$

$$S(s)=\frac{1}{1+L(s)}=\frac{1}{1+P(s)C(s)}$$

$$L(s)\to 0 \Leftrightarrow S(s)\to 1, T(s)\to 0$$

Zeros of L(s)

- **Sensitivity Value of 1 at all Uncancelled Plant Zeros (所有未消去的系统零点处的灵敏度值为1)**
  - No command following or output disturbance rejection available at frequencies corresponding to the pure imaginary plant zeros (在纯虚数系统零点对应的频率上不可用命令跟随或输出干扰抑制)

Effect of Plant Poles and Zeros

- **Zero Sensitivity Value at all Uncancelled Plant/Controller Poles** **(所有未消去的系统/控制器极点的灵敏度为零)**
  - The zeros of the sensitivity function include all uncancelled plant and controller poles **(灵敏度函数的零点包括所有未消掉的系统和控制器极点)**

$$L(s) \to \infty \Leftrightarrow S(s) \to 0, T(s) \to 1$$

Poles of L(s)

- **Complementary Sensitivity Value of 1 at all Uncancelled Plant and Controller Poles** **(所有未消掉的系统和控制器极点上的互补灵敏度值为1)**
  - Perfect command following and disturbance rejection at frequencies corresponding to pure imaginary poles of plant and controller **(在被控系统和控制器的纯虚数极点对应的频率段内有较好的指令跟踪和干扰抑制）**

$$Y(s) = T(s)F(s)R(s) + S_i(s)d_i(s) + S(s)d_0(s) - T(s)n(s)$$

Effect of Open Loop Integrators

- **Lemma 1 (System Types for <u>Reference Command</u>)** **(引理1(用于参考命令的系统类型))**

  For *stable* CL system with one-DOF controller configuration, assume that open-loop TF has i free integrators or i poles at the origin, i.e.,**(对采用单自由度控制器的稳定闭环系统，假设开环TF在原点处有*i*个自由积分器或*i*个极点，即)**

$$D_p(s)D_c(s) = s^i \bar{D}_p(s)\bar{D}_c(s), \quad i \geq 1$$

$$\bar{D}_p(0)\bar{D}_c(0) = c_0 \neq 0$$

$$N_p(0)N_c(0) = c_1 \neq 0$$

Then, for a step output disturbance or step reference command, the error $e(t)$ satisfies **(那么，对于阶跃输出干扰或阶跃参考命令，误差*e*(*t*)满足)**

$$\lim_{t\to\infty} e(t) = 0, \quad \forall i \geq 1$$

F.V.T

$$\int_0^\infty e(t)dt = 0, \quad \forall i \geq 2$$

Effect of Open Loop Integrators

- **Lemma 1 (cts)** **(引理1)**

  Also, for a negative unit ramp output disturbance or a positive unit ramp reference command, the error $e(t)$ satisfies **(此外，对于负的单位斜坡输出干扰或正的单位斜坡参考命令，误差*e*(*t*)满足)**

$$\lim_{t\to\infty} e(t) = \frac{c_0}{c_1}, \qquad \text{for} \quad i = 1$$

$$\lim_{t\to\infty} e(t) = 0, \qquad \forall i \geq 2$$

$$\int_0^\infty e(t)dt = 0, \qquad \forall i \geq 3$$

$$E(s) = [1 - T(s)F(s)]R(s) - S_i(s)d_i(s) - S(s)d_0(s) + T(s)n(s)$$

Effect of Open Loop Integrators

Q: *Is it possible to design a one-DOF controller to reject a ramp output disturbance that also has the property that the step response does not have any overshoot ?* (是否有可能设计一个单自由度控制器来抑制一个斜坡输出干扰，同时使阶跃响应没有任何超调的特性)

unit ramp reference $\lim_{t\to\infty} e(t) = 0$,

require $i \geq 2$, $\Rightarrow$ unit step: $\int_0^\infty e(t) = 0$

$$\Rightarrow \begin{cases} e(t) = r(t) - y(t) \\ e(0) = 1 - 0 = 1 \\ \int_0^t e(t)dt = 0 \end{cases} \Rightarrow$$

$e(t)$ has to be negative in some intervals ($e(t)$在某些时段必须为负值)

$\Rightarrow$ Overshoot! (超调)

Effect of Open Loop Integrators

- **Lemma 2 (System Types for <u>Input Disturbance</u>)** **(引理2(用于输入干扰的系统类型)**

For *stable* CL system with one-DOF controller configuration, assume that the controller TF has i free integrators or i poles at the origin, i.e., **(对于采用单自由度控制器的稳定闭环系统，假设控制器传递函数在原点处有*i*个自由积分器或*i*个极点，即)**

$$D_c(s) = s^i \bar{D}_c(s), \quad \bar{D}_c(0) = c_0 \neq 0, \quad N_c(0) = c_1 \neq 0$$

Then, for a step input disturbance, the error $e(t)$ satisfies **(对于阶跃输入干扰，误差*e(t)*满足)**

$$\lim_{t\to\infty} e(t) = 0, \qquad \forall i \geq 1$$

$$\int_0^\infty e(t)dt = 0, \qquad \forall i \geq 2$$

Also, for a negative unit ramp input disturbance, the error $e(t)$ satisfies **(对于负单位斜坡输入干扰，误差*e(t)*满足)**

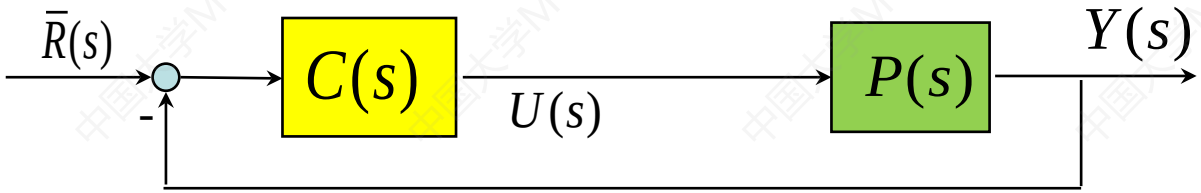
$$\lim_{t\to\infty} e(t) = \frac{c_0}{c_1}, \qquad \text{for} \quad i = 1$$

$$\lim_{t\to\infty} e(t) = 0, \qquad \forall i \geq 2$$

$$\int_0^\infty e(t)dt = 0, \qquad \forall i \geq 3$$

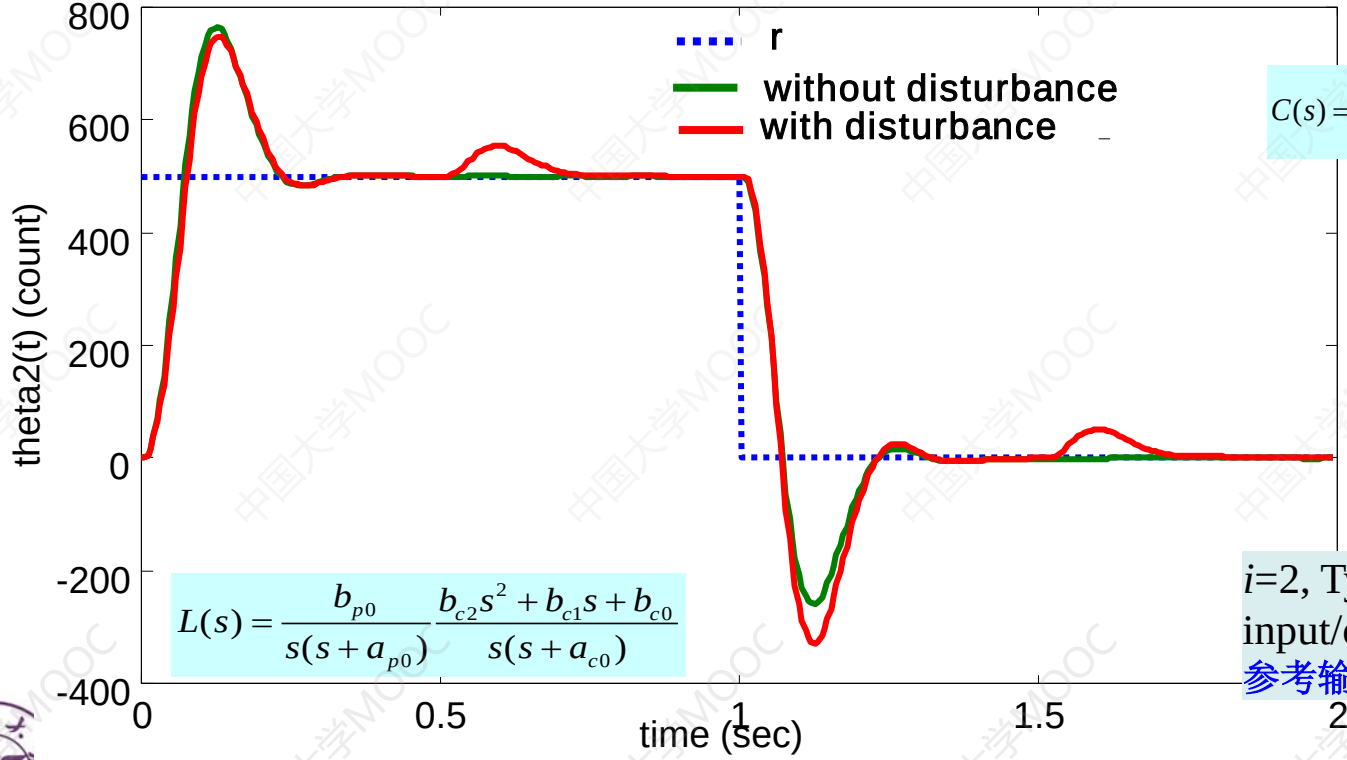
Effect of Open Loop Integrators

- **Remarks**
  - The above conclusion holds for a one-DOF feedback control system. Later we show that overshoot of step responses can actually be avoided if the architecture is changed to a two-DOF control system by using suitable pre-filter TF $F(s)$ (上述结论适用于单自由度反馈控制系统。如果通过采用适当的预滤波器传递函数$F(s)$将系统结构改为二自由度控制系统，实际上可以避免阶跃响应的超调)



PID Control Experimental Results

PID1 Step Response with/without Step disturbance (有/没有阶跃干扰的PID1阶跃响应)



$$L(s) = \frac{b_{p0}}{s(s+a_{p0})}\frac{b_{c2}s^2 + b_{c1}s + b_{c0}}{s(s+a_{c0})}$$

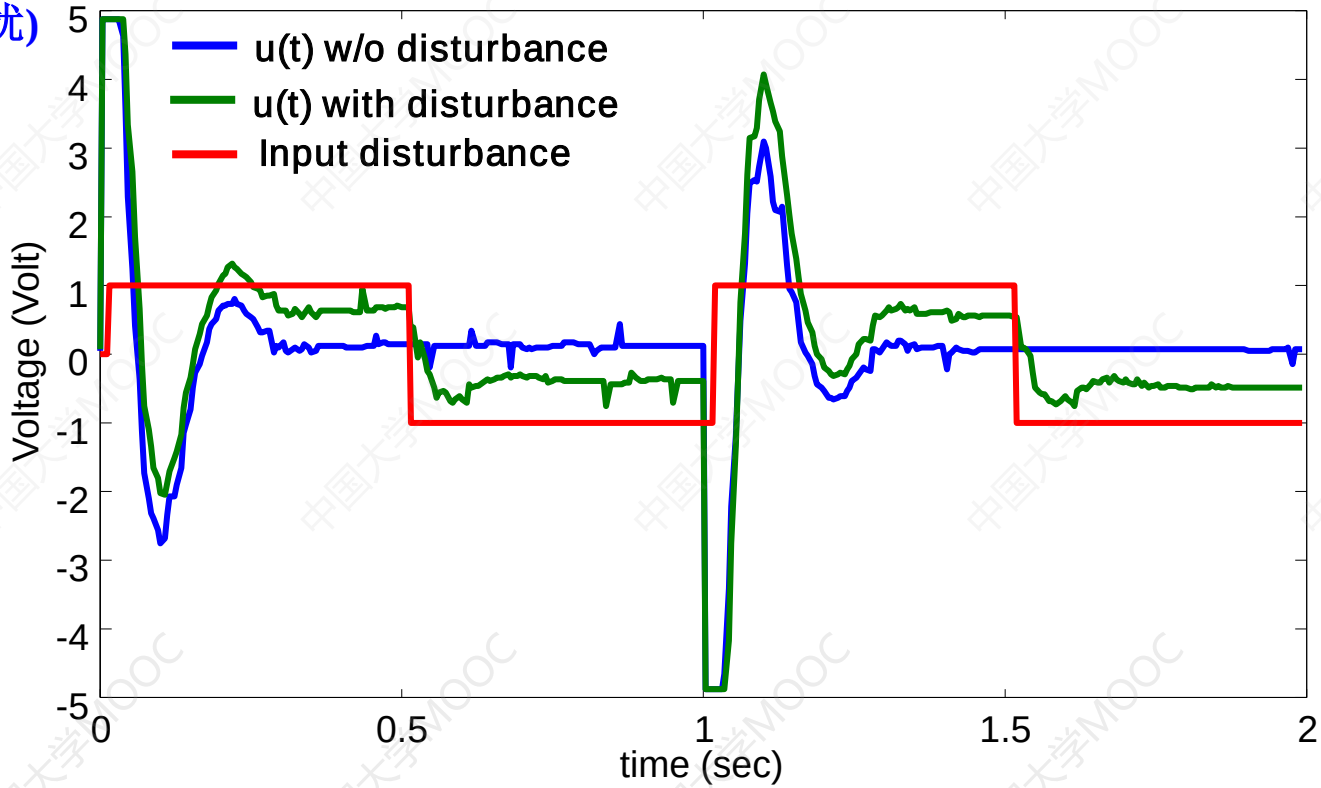
$$C(s) = \frac{(K_D + K_P\tau_D)s^2 + (K_P + K_I\tau_D)s + K_I}{\tau_D s^2 + s}$$

Type 1 w.r.t. input disturbance (相对于输入干扰为1型系统)

i=2, Type 2 w.r.t. reference input/output disturbances (相对于参考输入/输出干扰为2型系统)

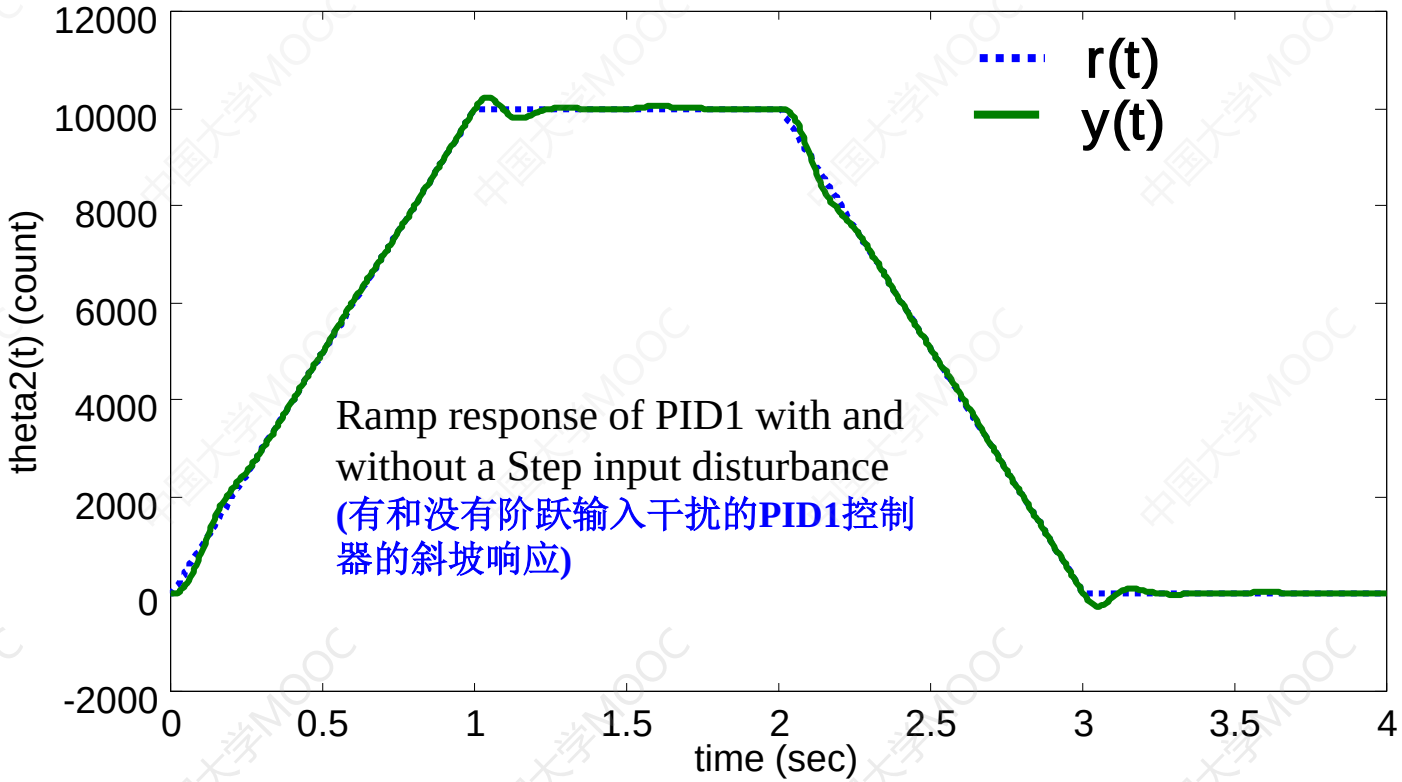
PID Control Experimental Results

Control Effort and Input Disturbance of PID1 Step Response (PID1阶跃响应的控制量和输入干扰)



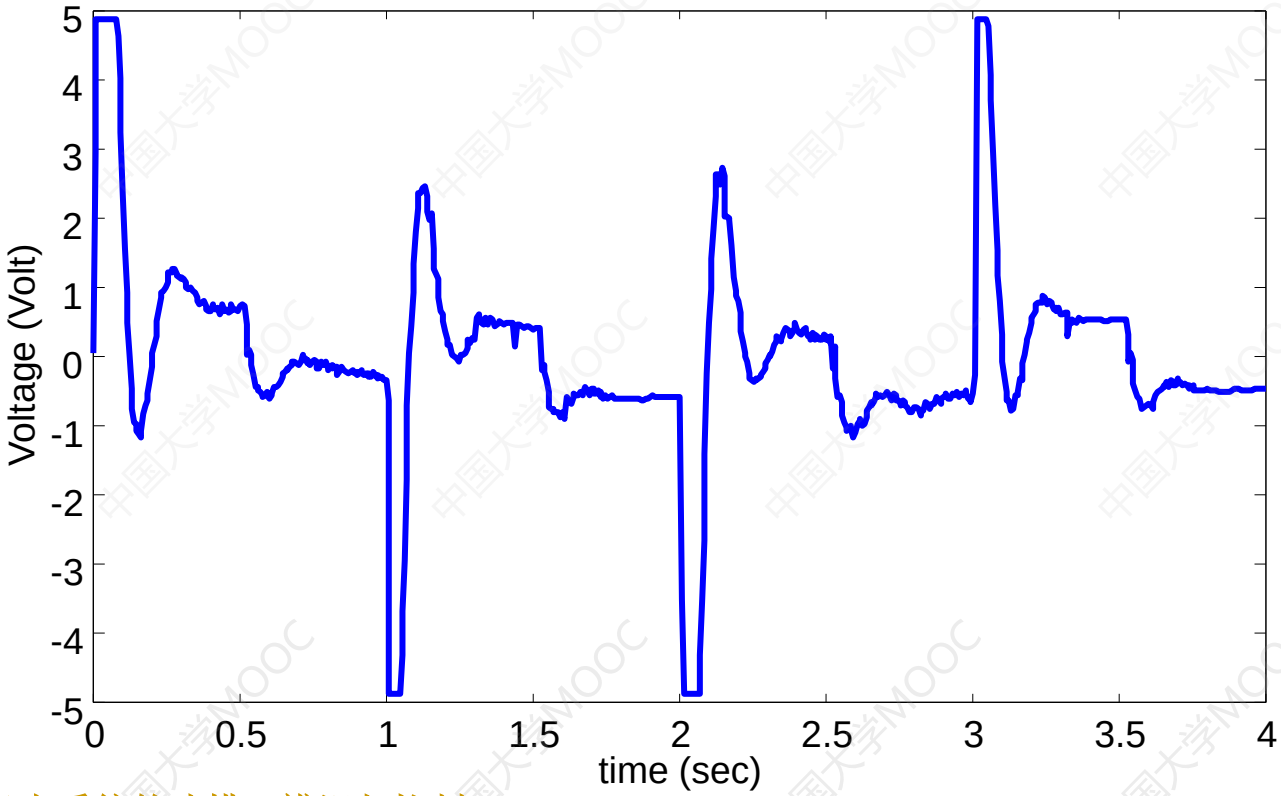
PID Control Experimental Results

PID1 Ramp Response with Disturbance (有干扰的PID1斜坡响应)



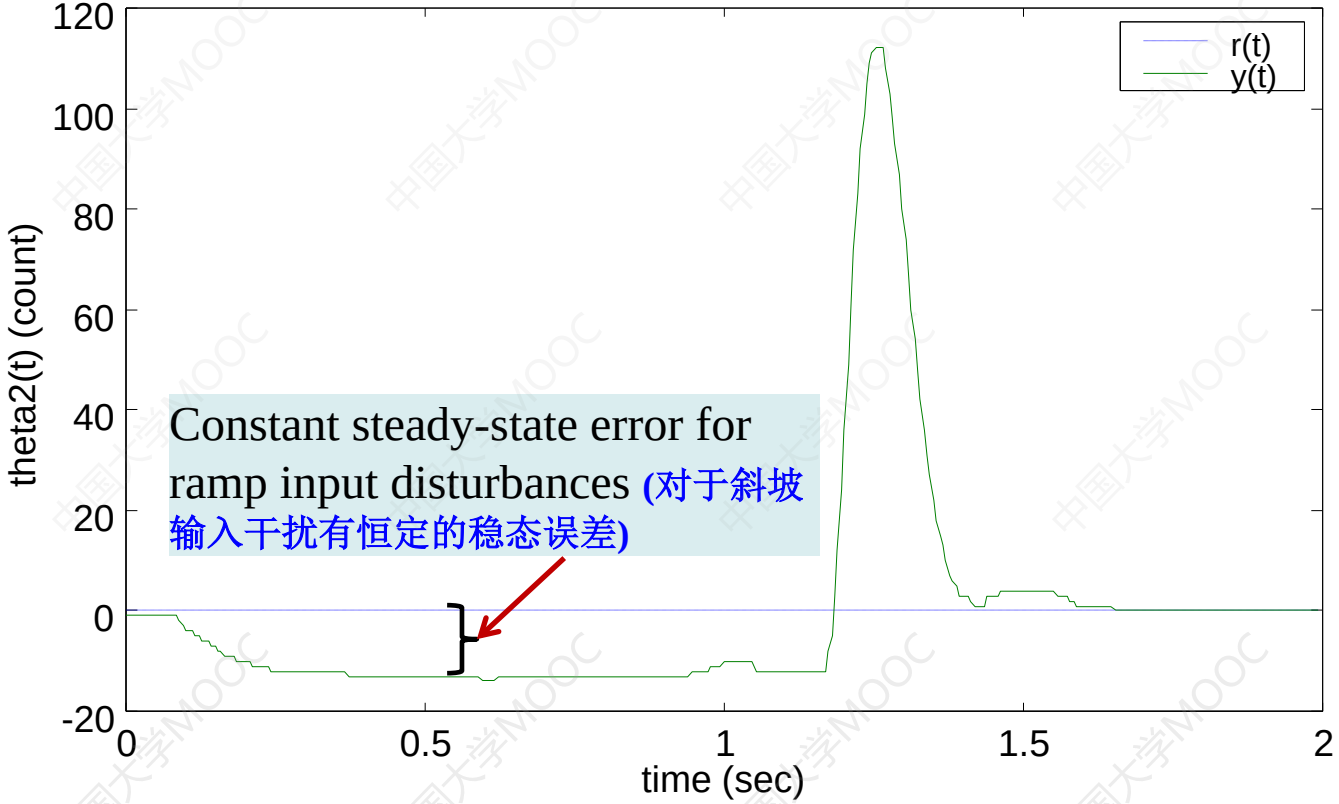
PID Control Experimental Results

Control Effort of PID1 <u>Ramp Response</u> (斜坡响应的PID1控制量)



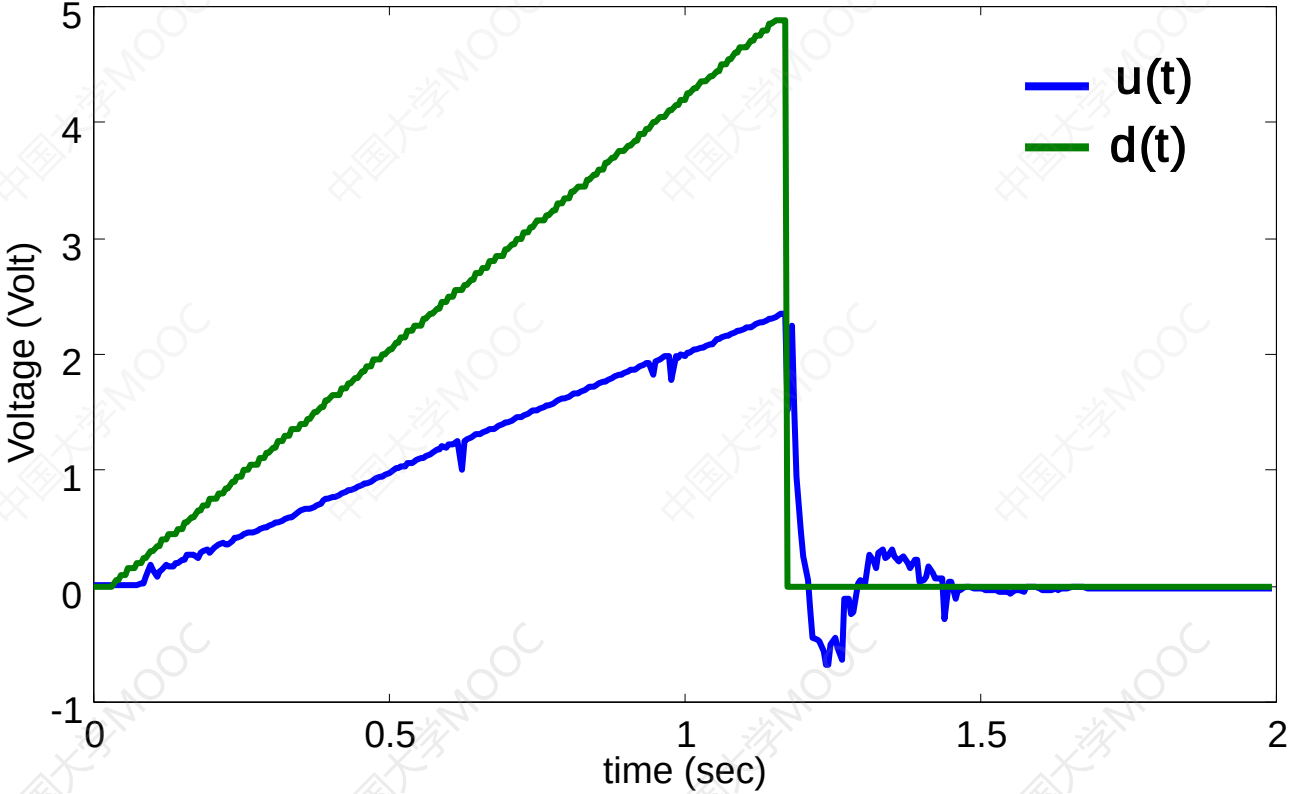
PID Control Experimental Results

PID1 Response with <u>Ramp Input Disturbance</u> (有斜坡输入干扰的PID1响应)



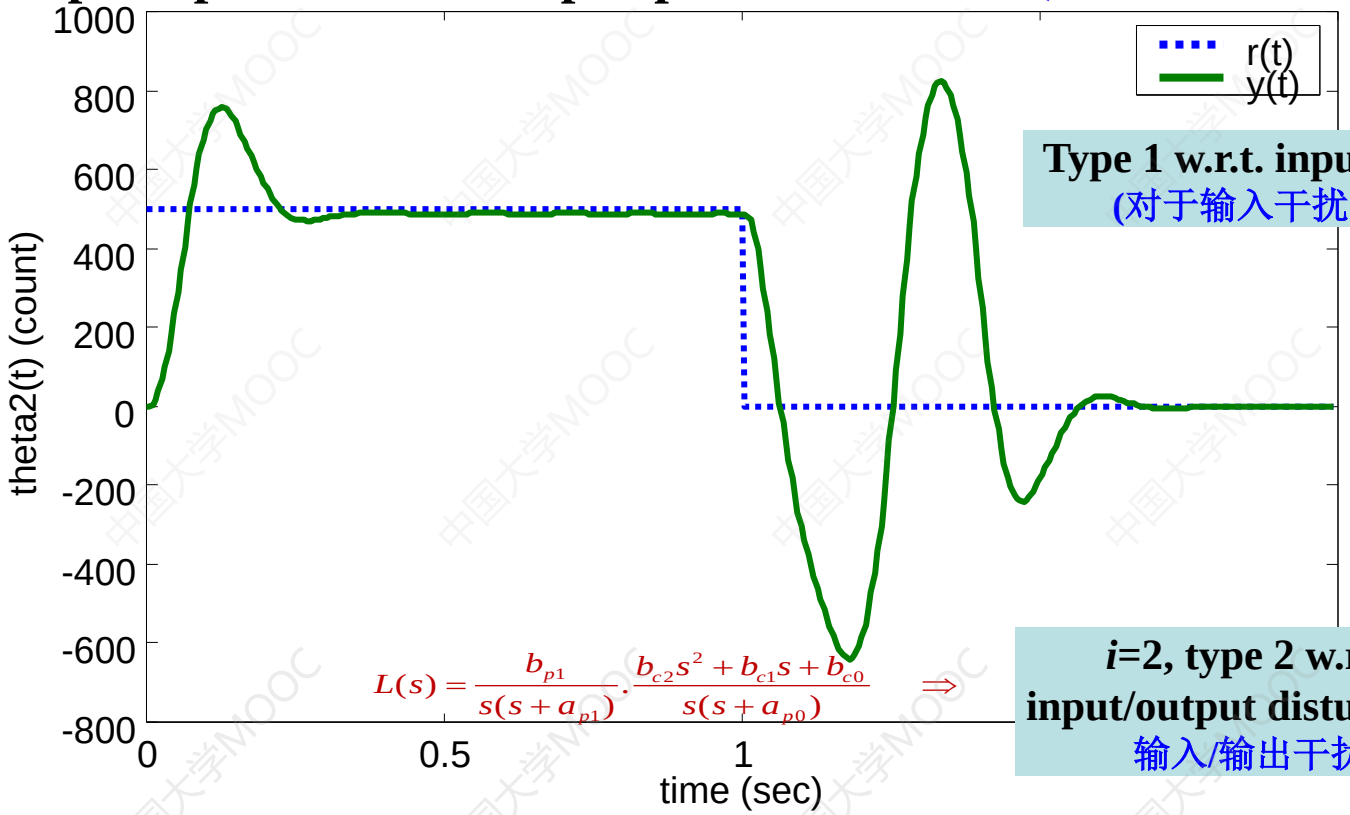
PID Control Experimental Results

Control Effort and Disturbance (控制量和干扰量)



PID Control Experimental Results

PID1 Step Response with Ramp Input Disturbance (斜坡输入干扰的PID1阶跃响应)



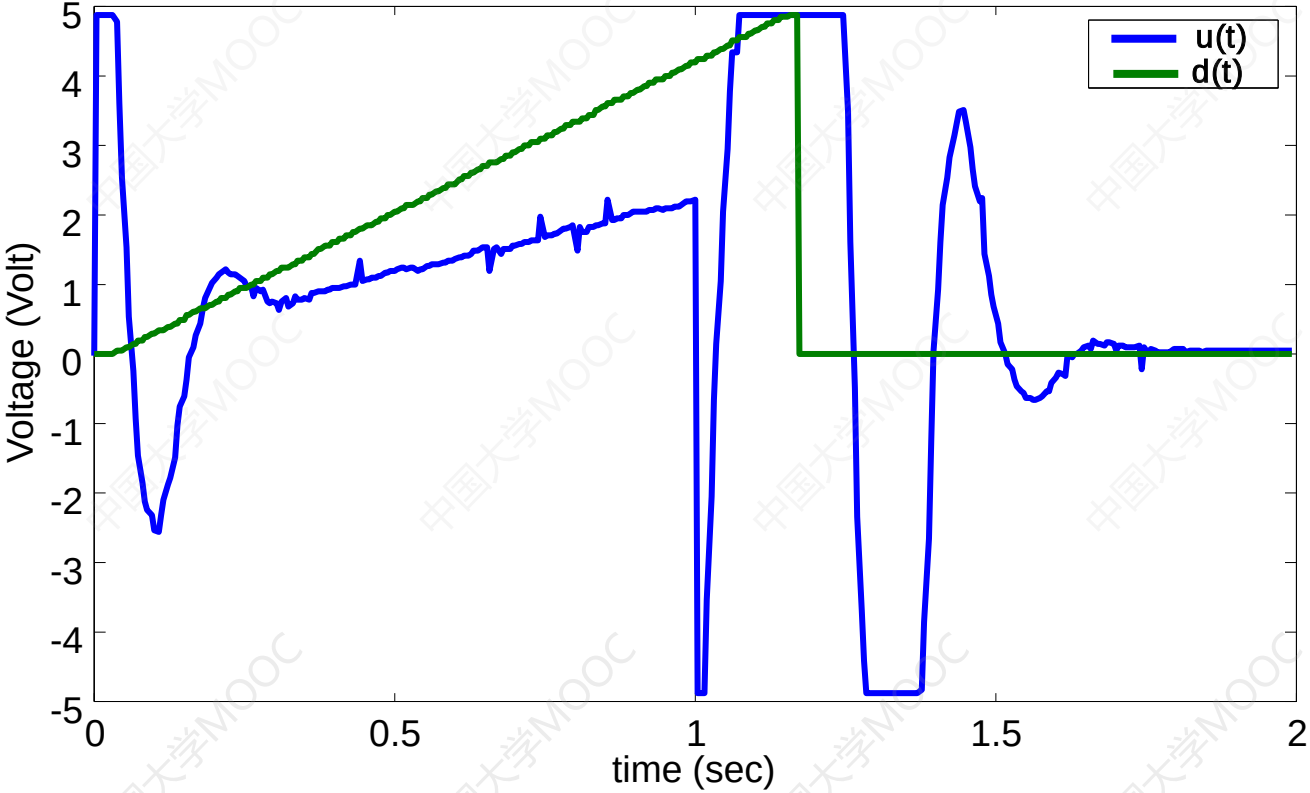
$$L(s) = \frac{b_{p1}}{s(s+a_{p1})} \cdot \frac{b_{c2}s^2 + b_{c1}s + b_{c0}}{s(s+a_{p0})} \Rightarrow$$

Type 1 w.r.t. input disturbance (对于输入干扰是1型系统)

***i*=2, type 2 w.r.t. reference input/output disturbance (对于参考输入/输出干扰是2型系统)**

PID Control Experimental Results

Control Effort and Disturbance (控制量和干扰量)



Summary

- **Effect of Plant Structure (结构影响)**
  - Time Delay (时延)
  - Plant Zeros and Poles (系统零极点分布)
  - Effect of Open Loop Integrators (开环积分的影响)